Mary C. Banner

GIS AND ENVIRONMENTAL SCIENCE PROFESSIONAL

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PROFESSIONAL SUMMARY

Environmental science and GIS professional with a master's degree and hands-on experience in drinking-water sampling, field measurements, environmental monitoring, and water-system operations. Experienced in groundwater sampling, pH, temperature and chlorine testing, macroinvertebrate monitoring, stream and habitat assessments, regulatory documentation, data QA/QC, and ArcGIS Pro mapping. Brings emergency-response and infrastructure-inspection experience from a community water utility, along with nearly a decade of public-facing work at Point Defiance Zoo & Aquarium. Offers accurate documentation, clear communication, and dependable independent fieldwork in changing outdoor conditions.

CORE QUALIFICATIONS

- Water-quality sampling and field measurements
- Groundwater wells and drinking-water systems
- Environmental monitoring and field documentation
- QA/QC and regulatory recordkeeping
- Macroinvertebrate and aquatic habitat data
- Infrastructure inspection and emergency response
- ArcGIS Pro, ArcGIS Online and StoryMaps
- Technical reports and public communication

PROFESSIONAL EXPERIENCE

NASA Arctic Boreal Vulnerability Experiment and SSAI | Remote — **Dec 2024 - Apr 2025**, **Apr 2026 - Aug 2026**

Graduate Research Intern and Scientific Communications Specialist (Part-Time)

- Integrated and reviewed remote-sensing, climate, hydrologic, ecosystem, and GIS datasets for regional environmental products covering Alaska and northwestern Canada.
- Completed QA/QC by checking sources, units, attributes, coordinate systems, geographic alignment, and project documentation before maps and datasets moved through scientific review.
- Produced community-specific Changing Northern Landscapes maps and plain-language booklets and coordinated revisions with scientists to preserve technical accuracy and clearly communicate limitations.

Fox Island Mutual Water Association | Fox Island, WA — **Jul 2025 - Apr 2026**

Water System Operator in Training

- Collected groundwater and drinking-water compliance samples and performed field measurements including pH, temperature, and chlorine residual; supported sampling for iron, manganese, nitrogen, and bacteriological parameters.
- Inspected wells, pumps, storage facilities, pressure systems, hydrants, valves, pressure-reducing valves, and meters; documented operating conditions and maintained accurate paper and regulatory records.
- Performed distribution-system flushing, valve exercising, meter reading, hydrant maintenance, leak investigations, and emergency repair support while working outdoors in changing conditions.
- Responded to customer concerns and service disruptions, coordinated with contractors and utility staff, and supported work performed under Washington State drinking-water requirements.

NASA Ames Indigenous Peoples Initiative | Remote — **Jun 2023 - Aug 2023**

Environmental Research Intern

- Authored a technical white paper on the use of NASA Earth-observation data for environmental awareness, watershed planning, and decision-making in Great Lakes Indigenous communities.
- Reviewed Tribal GIS initiatives and airborne, in-situ, and satellite data sources and applied Two-Eyed Seeing to connect Indigenous knowledge with geospatial tools and community priorities.

NASA Arctic Boreal Vulnerability Experiment | Remote — **Jun 2022 - Aug 2022**

Geospatial Research Intern

- Developed an ArcGIS StoryMap synthesizing environmental research across six Arctic-boreal study domains and translated complex findings into an accessible public-facing format.
- Integrated satellite, airborne, and field observations to communicate landscape, hydrologic, and ecosystem change to researchers, partner organizations, and community stakeholders.

Salish Sea Research Center and NASA Ames Research Center | Remote — **Jul 2021 - Aug 2021**

Environmental GIS Intern

- Built GIS datasets to evaluate pipeline crossings and potential impacts to watersheds, freshwater habitat, and culturally important manoomin ecosystems in Minnesota.
- Analyzed ecological risk and presented findings through a scientific poster focused on environmental protection, Indigenous food sovereignty, and community-relevant research.

Yakama Nation Fisheries | Cle Elum, WA **2020 - 2021**

Fisheries and Environmental Monitoring Intern

- Collected macroinvertebrate and environmental monitoring data used to evaluate aquatic ecosystem health and supported stream and habitat assessments, fish surveys, tagging, and field documentation.
- Assisted salmonid rearing, Sockeye reintroduction, Bull Trout rescue, fish transport, DNA sampling, and restoration work while following field safety and biological handling procedures.

Point Defiance Zoo & Aquarium | Tacoma, WA **Apr 2011 - Mar 2020**

Part-Time/Seasonal Employee

- Maintained grounds and public facilities through mowing, pruning, litter removal, sanitation, painting, and event preparation, helping keep the site safe, clean, and operational for staff, animals, and visitors.
- Supported Zoolights installation and teardown, converted displays to LED systems, repaired rope lighting, operated forklifts and high lifts, coordinated parking-lot activities, and secured gates and lighting systems.

EDUCATION

Master of Professional Studies in Environmental Science **May 2025**

SUNY College of Environmental Science and Forestry | Ecosystems Land Water and Air

Bachelor of Science in Native Environmental Science **Jun 2022**

Northwest Indian College | GIS focus

Associate of Arts in Sociology **Dec 2019**

Pierce College

TECHNICAL SKILLS AND CREDENTIALS

Field and water: Water sampling, handheld field measurements, groundwater wells, distribution systems, meter reading, system flushing, valves, hydrants, pumps, leak response, field logs, environmental surveys, and macroinvertebrate collection.

Data and mapping: ArcGIS Pro, ArcGIS Online, StoryMaps, Survey123, remote sensing, spatial analysis, cartography, metadata, Microsoft Excel, Word, PowerPoint, Adobe Creative Cloud, and Google Earth Engine training.

Credentials: Valid Washington State driver's license; Washington State Flagger Certification, January 2026 - January 2029; introductory backflow and cross-connection training.